Winter term 2020/2021: Exercises "Radio Frequency Technology"

EXERCISE SESSION 7 (18.12.2020)

Problem 7.1: Circular waveguide

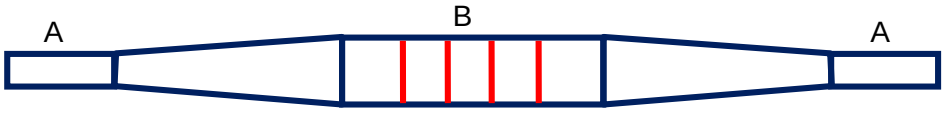
Consider an air-filled rectangular waveguide of width *a* and height *a/2* with a cutoff frequency (of the fundamental mode) of 10 GHz.

a) What is the upper frequency limit of the monomode frequency range of the rectangular waveguide?
b) For a circular cylindrical waveguide with the same area of the cross-section (as the above rectangular waveguide), calculate the lower and upper limits of the monomode frequency range.
c) For a circular cylindrical waveguide with the same circumference of the cross-section (as the above rectangular waveguide), calculate the lower and upper limits of the monomode frequency range.

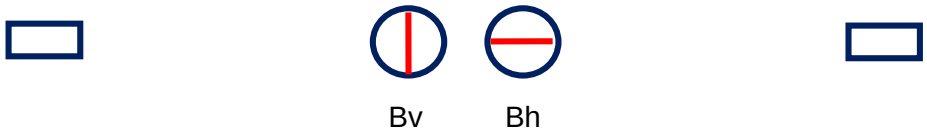
Problem 7.2: Field vectors

A mechanically variable attenuator in rectangular waveguide can be constructed as follows:

longitudinal view:



cross-sectional views:



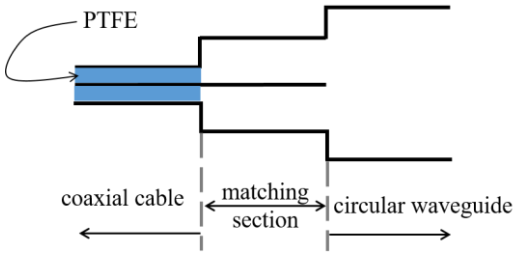
Here, a long gradual taper realizes the transition from rectangular cross section, A, to circular cross section, B. In the B-section, there are thin lossy wires placed periodically. Using a mechanism, the orientation of all these wires can be gradually changed from vertical, "Bv", to horizontal, "Bh".

a) Depict the electric field (in both intensity and vectorial direction) in the cross section of parts "A".
b) Depict the electric field (in both intensity and vectorial direction) in the cross section of part "B", thereby neglecting the effect of the red wires.
c) Considering the two respective orientations of the wires, "Bv" and "Bh", describe the effect of the wires on the attenuation (transmission loss) of the entire setup.

Problem 7.3: Circular waveguide TE mode excitation

A technique to excite TM_{01} mode in air-filled circular metal waveguide of diameter 14 mm from a coaxial cable of characteristics impedance 50 Ω is shown below. To achieve a good matching, an intermediate quarter-wavelength matching element (air filled coaxial section) between 50 Ω coaxial cable and metal waveguide is introduced. Ignore any power loss in the whole circuit design.

Winter term 2020/2021: Exercises "Radio Frequency Technology"



(a) Calculate the diameter of the inner pin of the coaxial cable, if the PTFE dielectric has dielectric constant of 2.1 and the inner dimeter of the outer conductor is 3.35 mm.
(b) What is the maximum frequency one could use this coaxial cable? Explain why.
(c) Determine the wave impedance of the TM_{01} mode in the circular metal waveguide at 20 GHz.
(d) Now calculate the inner diameter of the outer conductor and length of the matching coaxial section to achieve a perfect matching at 20 GHz.
(e) Assuming no radial component of magnetic field gets excited inside the circular waveguide, what is the cut-off frequency of the next mode in the circular waveguide that will get excited in this configuration? Also, draw the H-field along the circular cross-section for this mode.

Problem 7.4: Dispersion in circular waveguides

An air-filled hollow circular waveguide of radius 1 cm is 150 m long and is capped at the end with a metal plate. If a short pulse of frequency 19 GHz is introduced into the input end of the guide, multiple reflected pulses return back to the input side. Find the number of unique reflected pulses that will be detected and calculate their corresponding delays from the time of launching?

Short questions session 7

a) Which is the fundamental mode of a circular waveguide?

b) How should you modify the circular waveguide dimensions to increase the relative monomode bandwidth (i.e., bandwidth with respect to fundamental mode)?